

SAHA

Safeguarding Communities with Advanced Rainfall

analysis
Flood Prediction



AGENDA

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OUR TEAM



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INTRODUCTION

Disaster response encompasses a multitude of critical elements ranging from warning systems to immediate assistance. Among these, the warning system stands as a pivotal component, particularly in scenarios prone to flooding.

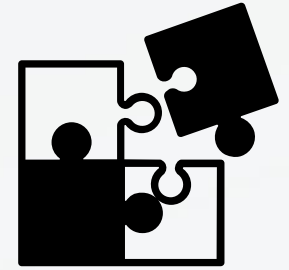
Recognizing the significance of early intervention, our team has dedicated efforts to enhance flood prediction mechanisms. Our solution, Sahay, not only offers a sophisticated warning system but also extends a user-friendly interface accessible to the general public. In this presentation, we delve into the core features of Sahay, highlighting its feature. Moreover, we introduce a proactive notification mechanism that alerts users of potential flood risks based on river conditions.

Communities can prepare effectively, minimize loss, and swiftly alert local populations to mitigate the impact of impending floods.





PROBLEM STATEMENT



- Despite advancements in technology and scientific understanding, flood prediction and management remain challenging due to traditional methods inaccuracy and lack of real-time data.
- Flood prediction and disaster management websites are needed to address outdated, inaccurate, and lack of real-time data flood prediction methods. This comprehensive system can provide accurate predictions, timely warnings, and effective disaster response strategies, reducing human and economic costs.

OBJECTIVES



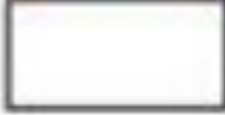
- The objectives of disaster management plan should be to localize a disaster, and to the maximum extent possible, contain it so as to minimize the impact on life, the environment and property.
- By analyzing various data sources, such as weather patterns, river levels, and historical flood data, our models can predict the likelihood and severity of floods in specific areas with high accuracy.





FLOOD AFFECTED AREAS IN INDIA (1998-2022)

Legend

-  Flood Affected Area
-  Normal River/Water Bodies
-  India Boundary



LITERATURE REVIEW

India, due to its diverse geographical features and heavy monsoon season, experiences frequent and devastating floods. Flood affected area is 39.76 Lakh Ha in Bihar due to floods in Ganga, Kosi rivers affecting 38 districts which is highest in the country. Flood affected area is 26.62 Lakh Ha, in Uttar Pradesh spread over 72 districts due to floods in Ganga river system every year, Flood affected area is 24.56 Lakh Ha in Assam which spread over 35 districts due to the floods that occur in Brahmaputra, Barak rivers almost every year. Flood affected area in these 3 states amounts to 60% of total flooded area in the country.

This project focuses on building a user-friendly and accessible flood prediction system. The system will integrate various elements:

- **Data Acquisition:** Real-time data on rainfall, river levels, and other relevant parameters will be collected from various sources.
- **Visualization and User Interface:** A user-friendly web-based platform will be developed to display real-time and historical data, flood forecasts, and risk maps. This platform will be accessible to the public and disaster management authorities.
- **Early Warning System:** An efficient early warning system will be established to disseminate timely and clear flood warnings to the public. This system will utilize pre-defined alert thresholds based on critical river levels.

. Expected Benefits

This flood prediction system is expected to provide significant benefits:

- **Reduced Loss of Life and Property:** By enabling early warnings and facilitating timely evacuation, the system can significantly reduce the number of casualties and the extent of property damage during floods.
- **Enhanced Preparedness:** The system will empower communities to prepare better for potential floods by providing them with advance information about flood risks and potential inundation areas.
- **Improved Decision-Making:** Disaster management authorities will have access to accurate flood forecasts and real-time data to make informed decisions regarding resource allocation, evacuation strategies, and rescue operations.

This flood prediction system is a crucial step towards mitigating the impact of floods. By providing timely warnings, facilitating preparedness, and empowering communities, the system has the potential to save lives, minimize property damage, and enhance overall flood resilience in the region.

PROPOSED WORK

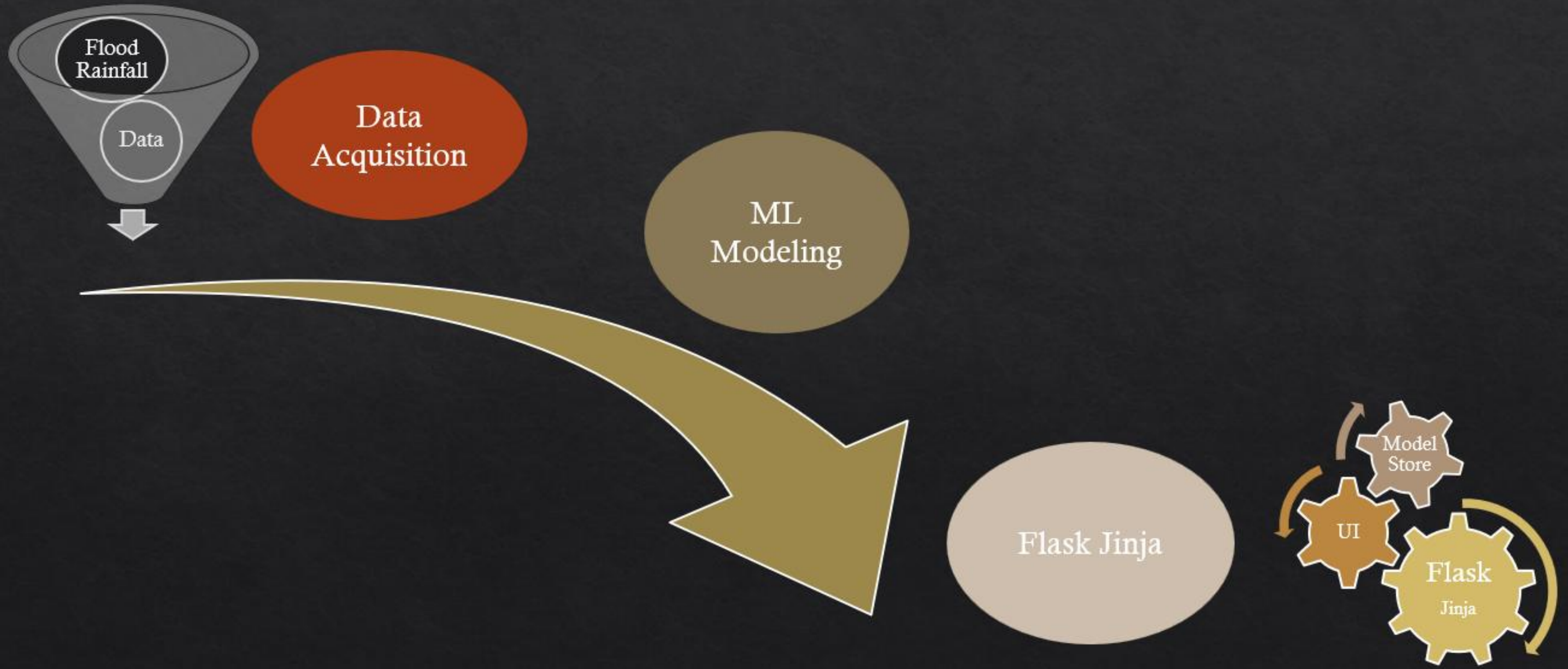


Flood Warning System to Minimize loss of life and property damage caused by floods through a proactive and user-friendly flood warning system.

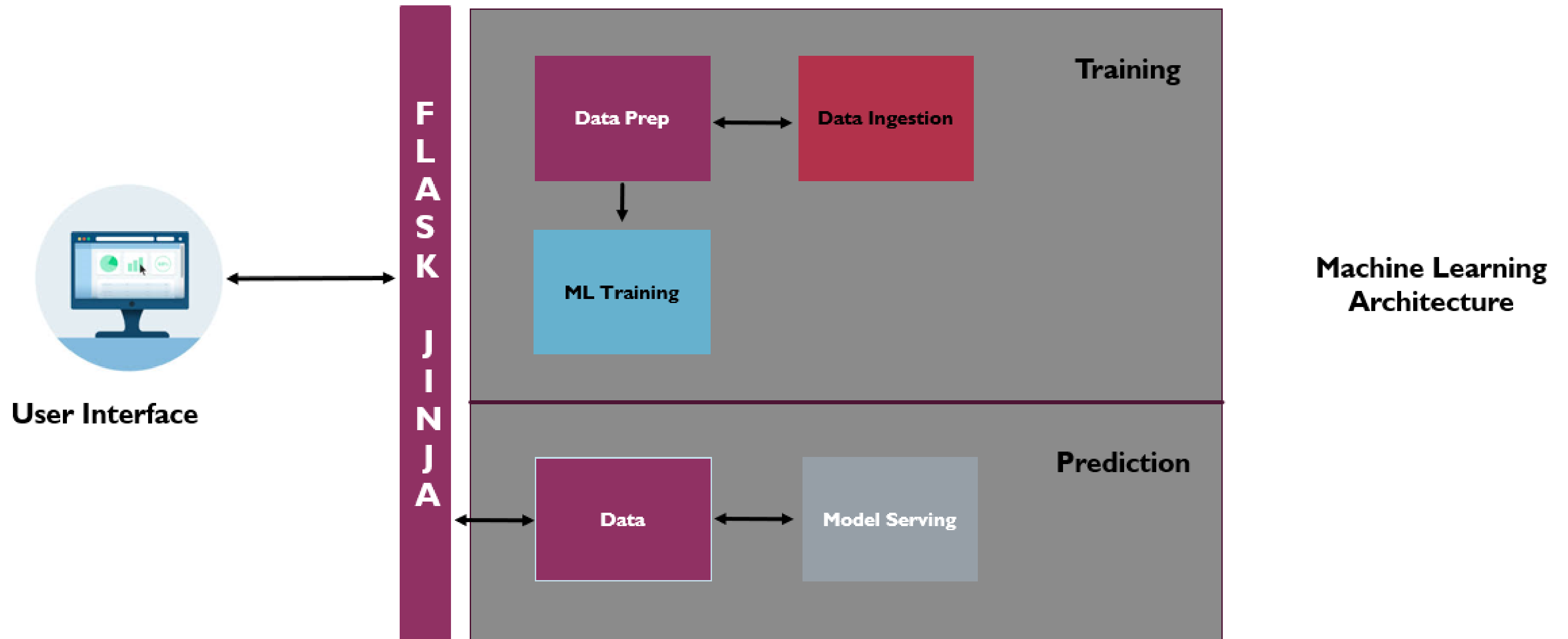
System Components:

- **Water Level Monitoring:** Monitoring of water flow levels in rivers.
- **Data Visualization:** Displaying insightful graphs.
- **Future Water Flow Predictions:** Information on projected water levels in rivers .
- **Flood Risk Notifications:** Automated alerts for potential flooding based on predicted water levels.
- **Historical River Flow Trends:** Visualization of past water flow data in rivers.
- **Sub-Divisional Rainfall Patterns:** Display of historical and potential rainfall patterns in the user's area.
- **Chatbot :** To provide assistance.

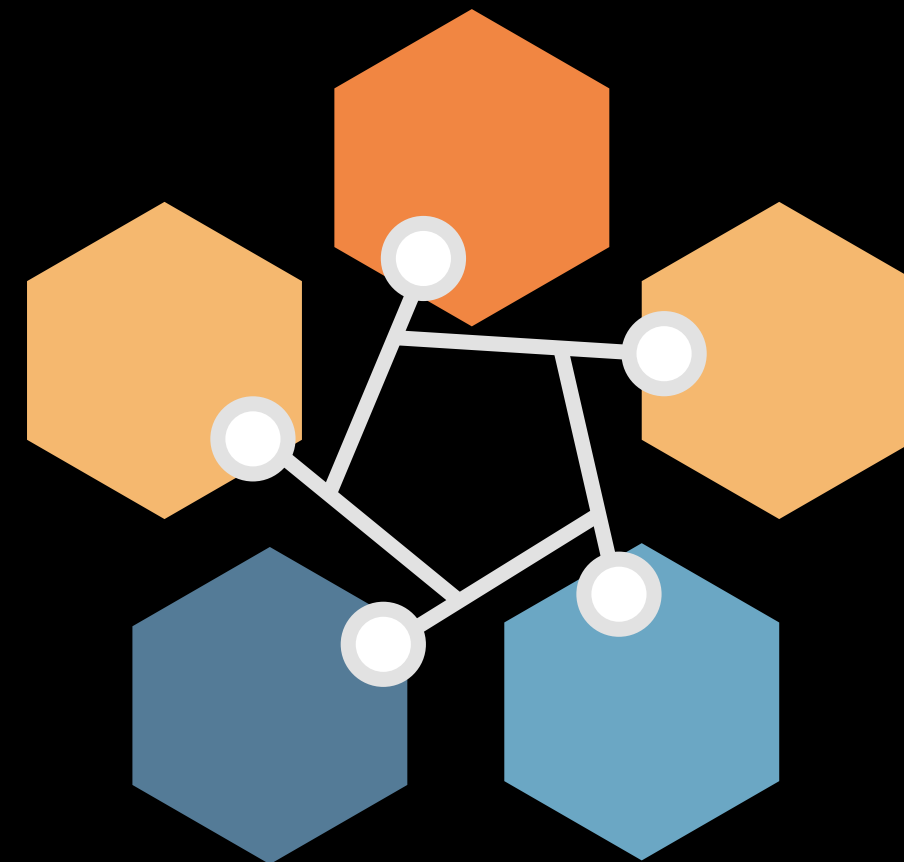
Solution approach

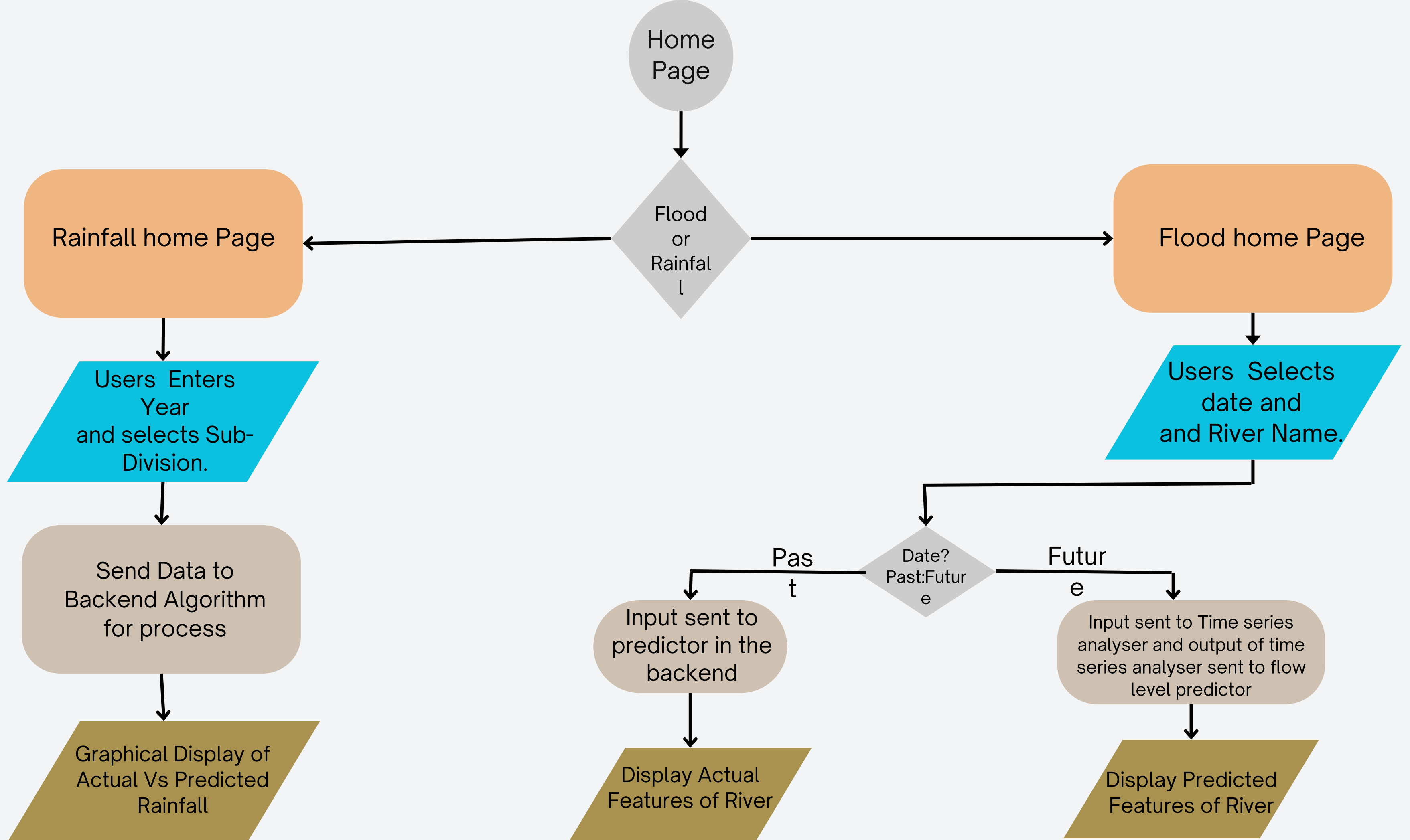


Architectural Diagram



Workflow Chart





Technologies

SOFTWARE REQUIREMENTS

01

Back-End
Python



02

Machine Learning
Pandas, NumPy,
Sklearn

03

Framework
Flask



04

Front-End
HTML, CSS
& JavaScript



Software Requirements

Python:

Description: Python is a high-level programming language known for its simplicity and readability. It offers vast libraries and frameworks for various purposes.

Specific Use: Python will be the primary language for developing the backend logic, machine learning algorithms, and other components of the web application.

Flask:

Description: Flask is a lightweight and versatile web framework for Python. It simplifies the process of building web applications by providing tools and libraries for handling requests, routing, and more.

Specific Use: Flask will be used to create the backend of the web application. It will handle HTTP requests, serve machine learning predictions, and render HTML templates.

HTML, CSS, JavaScript:

Description: HTML (Hypertext Markup Language), CSS (Cascading Style Sheets), and JavaScript are the core technologies used for building web pages and adding interactivity to web applications.

Specific Use: HTML will define the structure of the web pages, CSS will style them, and JavaScript will add interactive elements to enhance user experience in the frontend of the application.

Pandas:

Description: Pandas is a powerful data manipulation and analysis library for Python. It provides data structures and functions to efficiently work with structured data.

Specific Use: Pandas will be used for data preprocessing tasks such as cleaning, transforming, and organizing the flood-related data before feeding it into machine learning models.

NumPy:

Description: NumPy is a fundamental package for numerical computing in Python. It provides support for large arrays and matrices, along with mathematical functions to operate on them.

Specific Use: NumPy will complement Pandas by providing efficient numerical operations and array manipulation functions, especially when working with multidimensional data in the flood prediction model.

Scikit-learn:

Description: Scikit-learn is a popular machine learning library for Python. It offers simple and efficient tools for data mining and data analysis, including various algorithms for classification, regression, clustering, and more.

Specific Use: Scikit-learn will be used to build and train machine learning models for flood prediction based on historical data and relevant features.

Matplotlib and Seaborn:

Description: Matplotlib is a comprehensive library for creating static, interactive, and animated visualizations in Python. Seaborn is built on top of Matplotlib and provides a high-level interface for drawing attractive statistical graphics.

Specific Use: Matplotlib and Seaborn will be used to visualize the flood-related data, model predictions, and evaluation metrics, helping users gain insights into the data and communicate results effectively.

Visual Studio Code (VS Code):

Description: Visual Studio Code is a lightweight yet powerful source code editor developed by Microsoft. It supports various programming languages and extensions, making it highly customizable and suitable for various development tasks.

Specific Use: The Python code, HTML, CSS, and JavaScript files will be written, debugged, and managed using VS Code. It provides features like syntax highlighting, code completion, debugging, and version control integration to streamline the development workflow.

REAL TIME USAGE



Our project empowers individuals to understand flood risks and prepare accordingly.



Automated alerts for potential flooding based on predicted water levels.



Users can also see the historical trends of rivers flow and can visualize the rainfall patterns



PROJECT TIMELINE

We have established a meticulously crafted timeline, designed to align with the project's goals. Strict adherence of the said timeline will lead to the successful completion of the project within the set deadline.

Project Initiation



Review and
Editing

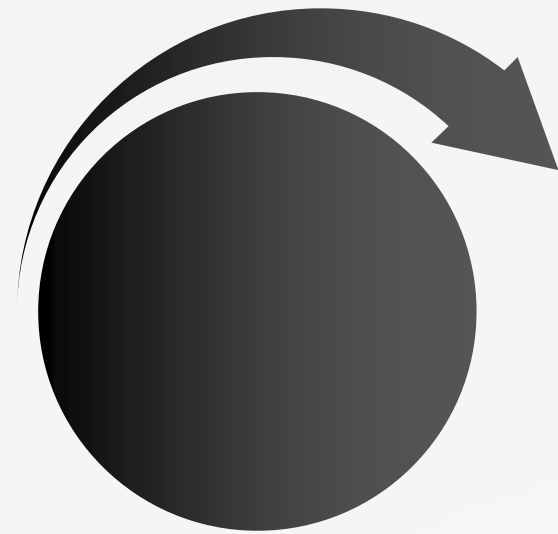


Presentation and
Sharing



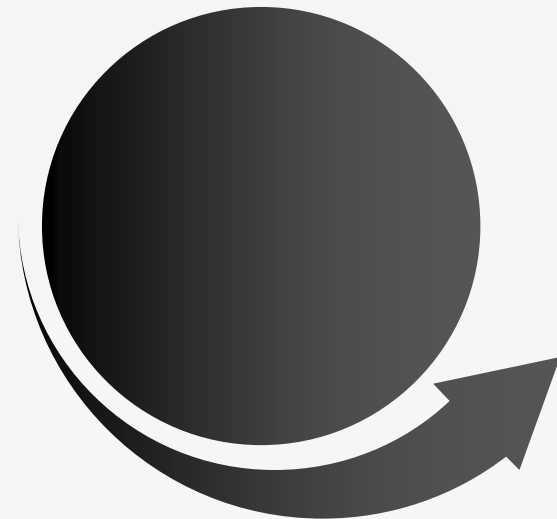
Week-1 to 4:
02 Mar - 27 Mar

Once the presentation is reviewed by the supervisor the team will start constructing the front-end of the project and gather the necessary data for the project.



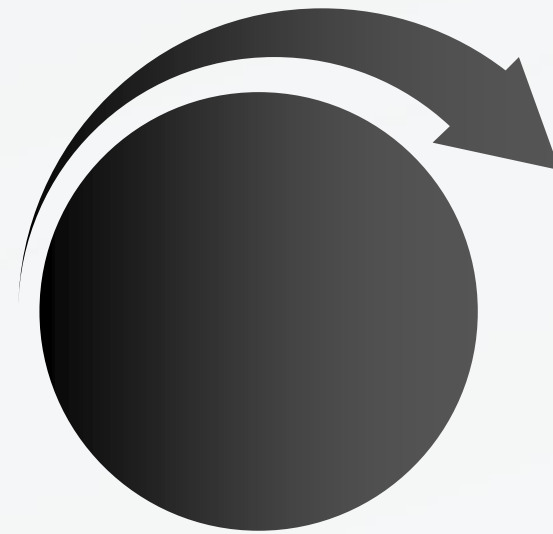
Project Review-1:
26 Feb - 01 Mar

Upon approval of the idea by the assigned project supervisor, the team will present a detailed presentation to the supervisor highlighting the crux of the project.



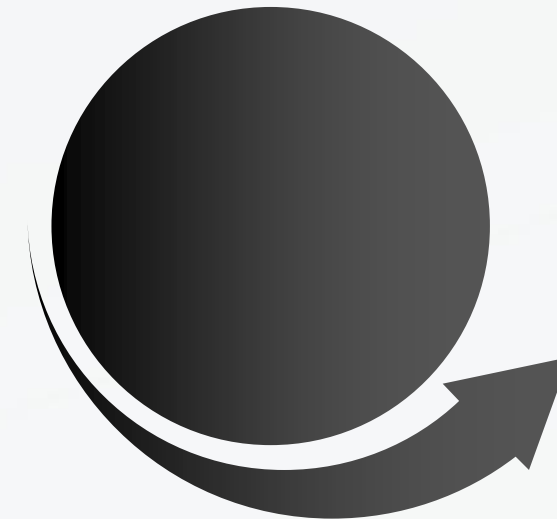
Project Review-2:
28 Mar - 03 Apr

The team will present a working frontend of the project to the supervisor with a few options for a reliable dataset.



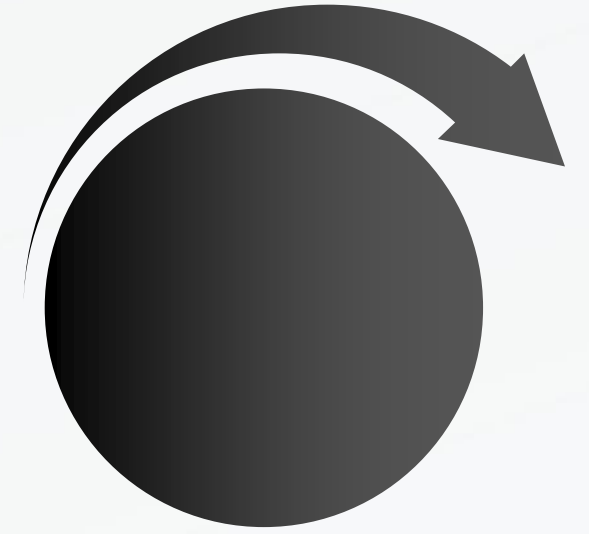
Week-5 to 7:
04 Apr- 21 Apr

Once the frontend of the project is positively reviewed by the project supervisor the team shall proceed with finalising the database for the project and begin the process of filtering the data.



Project Review-3:
22 Apr - 28 Apr

On the day of final review the team will present a working application to the supervisor with a reliable dataset.



References

- Central Water Commission: <https://ffs.india-water.gov.in/>
- Climate Hazards & Vulnerability Atlas of India: <https://imdpune.gov.in/hazardatlas/floodnew.html>
- National Disaster Management Authority, Govt of India: <https://ndma.gov.in/Natural-Hazards/Floods>
- Department of Water Resources, River Development and Ganga Rejuvenation : <https://jalshakti-dowr.gov.in/flood-forecasting/>

The background features abstract, wavy line patterns in the top-right and bottom-left corners, resembling topographical maps or fluid motion. The lines are thin and grey, creating a sense of depth and movement. The central text is bold and black, standing out against the white background.

**THANK
YOU**